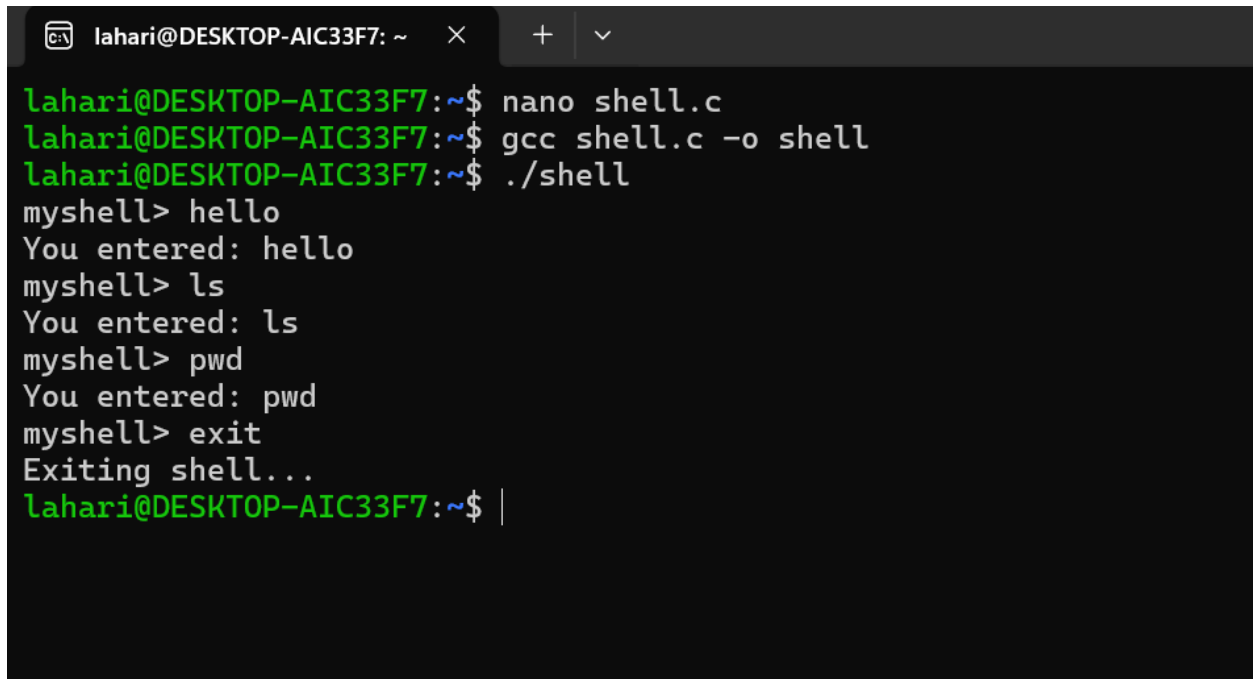


Skill-2

To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

A terminal window with a dark background and light green text. The window title bar shows 'lahari@DESKTOP-AIC33F7: ~' and standard window controls. The terminal shows the following sequence of commands and output:

```
lahari@DESKTOP-AIC33F7:~$ nano shell.c
lahari@DESKTOP-AIC33F7:~$ gcc shell.c -o shell
lahari@DESKTOP-AIC33F7:~$ ./shell
myshell> hello
You entered: hello
myshell> ls
You entered: ls
myshell> pwd
You entered: pwd
myshell> exit
Exiting shell...
lahari@DESKTOP-AIC33F7:~$ |
```

Program:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char command[100];
```

```
    while (1) {
```

```
        printf("myshell> ");
```

```
fgets(command, sizeof(command), stdin);
```

```
command[strcspn(command, "\n")] = '\0';
```

```
if (strcmp(command, "exit") == 0) {
```

```
    printf("Exiting shell...\n");
```

```
    break;
```

```
}
```

```
if (strlen(command) == 0) {
```

```
    continue;
```

```
}
```

```
printf("You entered: %s\n", command);
```

```
}
```

```
return 0;
```

```
}
```

```
lahari@DESKTOP-AIC33F7:~$ nano keyboard_shell.c
lahari@DESKTOP-AIC33F7:~$ gcc keyboard_shell.c -o keyboard_shell
lahari@DESKTOP-AIC33F7:~$ ./keyboard_shell

myshell> hello
Command entered: hello

myshell> testing123
Command entered: testing123

myshell> o
Command entered: o

myshell> hello
Command entered: help
                hello

myshell> exit
Exiting shell...
lahari@DESKTOP-AIC33F7:~$ |
```

Program:

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <termios.h>
```

```
#include <string.h>
```

```
#define SIZE 100
```

```
int main() {
```

```
    struct termios old, new;
```

```
    char buffer[SIZE];
```

```
    char ch;
```

```
    int index = 0;
```

```
// Get current terminal settings
tcgetattr(STDIN_FILENO, &old);
new = old;

// Read one character at a time
new.c_lflag &= ~(ICANON | ECHO);

tcsetattr(STDIN_FILENO, TCSANOW, &new);

while (1) {
    printf("\nmyshell> ");
    fflush(stdout);

    index = 0;

    while (1) {
        read(STDIN_FILENO, &ch, 1);

        // Enter
        if (ch == '\n' || ch == '\r') {
            buffer[index] = '\0';
            break;
        }
    }
}
```

```
// Backspace
if (ch == 127 || ch == 8) {
    if (index > 0) {
        index--;
        printf("\b \b");
        fflush(stdout);
    }
    continue;
}

// Normal character
if (index < SIZE - 1) {
    buffer[index] = ch;
    index++;

    printf("%c", ch);
    fflush(stdout);
}

}

// Exit
if (strcmp(buffer, "exit") == 0) {
    printf("\nExiting shell...\n");
}
```

```
        break;
    }

    // Empty input
    if (index == 0) {
        continue;
    }

    printf("\nCommand entered: %s\n", buffer);
}

// Restore terminal
tcsetattr(STDIN_FILENO, TCSANOW, &old);

return 0;
}
```